

Summary Report

Week 1-2: Data Cleaning & Exploratory Data Analysis — AnalystLab Africa Internship

1. Cleaning Challenges Encountered

Both datasets required distinct cleaning strategies, summarized below.

Dataset	Cleaning Challenge	Action Taken
Online Retail	~25% of rows missing CustomerID	Flagged (-1) instead of removed; excluded only from customer-level analysis
Online Retail	Negative quantities & cancelled invoices (prefix "C")	Separated into a distinct dataframe rather than deleted, to preserve return/cancellation data
Online Retail	5,268 duplicate rows; unit prices ≤ 0	Duplicates removed; invalid-price rows removed from the clean dataset
Netflix	3 rows had a duration value misplaced into the rating column	Values shifted back to the duration column; rating treated as missing
Netflix	duration column mixed two units (minutes for Movies, seasons for TV Shows)	Split into duration_value (numeric) and duration_unit (min / Season(s))
Netflix	Missing director, cast, country, rating, date_added	Text fields filled with "Unknown"; rating filled with mode; rows missing date_added/duration removed (small number)

2. Key EDA Findings

Online Retail

- **Market concentration:** the United Kingdom generates 84.6% of total revenue, with the remaining 15.4% spread across 37 other countries.
- **Seasonality:** monthly revenue rises sharply toward the end of the year, peaking in November 2011 at over £1.5M, consistent with holiday shopping demand.
- **Order size:** most order lines involve small quantities (under 15 units), with a long tail of bulk orders suggesting a mixed retail/wholesale customer base.

Netflix

- **Content mix:** Movies account for 69.7% of the catalog versus 30.3% for TV Shows.
- **Catalog growth:** titles added per year grew sharply from 2016 to a peak in 2019 (2,016 titles), then declined in 2020-2021, coinciding with pandemic-related production disruptions.
- **Origin & rating:** the United States is the leading content-producing country (3,683 titles), and TV-MA/TV-14 are the most common ratings, indicating a catalog weighted toward teen and adult audiences.

3. Top Insights

- **1.** A small subset of products and a single country (UK) drive a disproportionate share of Online Retail's sales, indicating high concentration risk alongside strong core-market performance.
- **2.** Seasonal demand spikes (Online Retail, Nov. 2011) and catalog growth patterns (Netflix, 2016-2019) both point to the value of trend-aware planning for stocking, staffing, or content acquisition.
- **3.** Data quality issues differed meaningfully between structured transactional data (missing IDs, cancellations, duplicates) and catalog metadata (misaligned fields, mixed units), showing that cleaning strategy must be tailored to each dataset's nature rather than applied generically.
- **4.** Approximately 25% of Online Retail transactions lack a CustomerID, limiting customer-level analysis (e.g. retention, lifetime value) on a meaningful portion of the data.
- **5.** Netflix's catalog composition (mostly Movies, mature-audience ratings, US-centric origin) reflects clear strategic patterns that could inform content acquisition or regional expansion decisions.